



## AQ8.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Consider two vectors  $a = (1, -2, 2)$ ,  $b = (4, 0, -3)$   $a = (1, -2, 2)$ ,  $b = (4, 0, -3)$ . Using the standard dot product as the inner product, the projection of  $a$  in the direction of  $b$  is given by

☐

$$\left(\frac{-8}{25}, 0, \frac{6}{25}\right)(25-8, 0, 256)$$

☐

$$\left(\frac{-8}{25}, \frac{6}{25}\right)(25-8, 256)$$

☐

$$\left(\frac{8}{25}, 0, \frac{-6}{25}\right)(258, 0, 25-6)$$

☐

$$\left(\frac{8}{25}, \frac{-6}{25}\right)(258, 25-6)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\left(\frac{-8}{25}, 0, \frac{6}{25}\right)(25-8, 0, 256)$$

1 point

Consider an orthonormal basis  $\left\{\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)\right\}$  of  $\mathbb{R}^2$ . If  $(3, 5) = c_1 v_1 + c_2 v_2$  where  $c_1, c_2 \in \mathbb{R}$  and

$$\frac{1}{\sqrt{5}}(1, 2), \quad \frac{1}{\sqrt{5}}(-2, 1)$$

of  $\mathbb{R}^2$ . If  $(3, 5) = c_1 v_1 + c_2 v_2$  where  $c_1, c_2 \in \mathbb{R}$  and  $v_1 = \frac{1}{\sqrt{5}}(1, 2)$ ,  $v_2 = \frac{1}{\sqrt{5}}(-2, 1)$  then choose the correct option.

$$\frac{1}{\sqrt{5}}(1, 2), \quad \frac{1}{\sqrt{5}}(-2, 1)$$

then choose the correct option.

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$$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}} c_1 = 5$$

$$13, c_2 = 5$$

$$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}} c_1 = -5$$

$$-13, c_2 = -5$$

$$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}} c_1 = -5$$



$$\sqrt{13}, c_2 = 5$$

$$\sqrt{-1}$$

$$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}c_1 = 5$$

$$\sqrt{-13}, c_2 = 5$$

$$\sqrt{-1}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}c_1 = -5$$

$$\sqrt{13}, c_2 = 5$$

$$\sqrt{-1}$$

*1 point*

Consider an orthonormal basis  $\{(1, 0, 0), (0, 1, 0)\}$  for a subspace  $W$  in  $R^3$ . If  $x = (1, 2, 3)$  is a vector in  $R^3$ , then which of the following represents a vector in  $W$  whose distance from  $x$  is the least? Consider dot product as the standard inner product.

$$\circ$$

$$(2, 4, 0)$$

$$\circ$$

$$(3, 4, 0)$$

$$\circ$$

$$(4, 5, 0)$$

$$\circ$$

$$(1, 2, 0)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$(1, 2, 0)$$

## Level 2

Consider the inner product  $\langle a, b \rangle = a_1 b_1 - a_1 b_2 - a_2 b_1 + 4a_2 b_2$  where  $a = (a_1, a_2)$ ,  $b = (b_1, b_2)$  are vectors in  $R^2$ . Use this information to answer questions 4 and 5.

*1 point*

Let  $x = (1, 2)$ . Find the projection of  $x$  in the direction of  $(3, 4)$  using the inner product defined above.

$$\circ$$

$$\frac{25}{49}(3, 4)$$

$$\circ$$

$$\frac{11}{25}(3, 4)$$

☐

$\frac{25}{49}(1, 2)$

☐

$\frac{11}{25}(1, 2)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\frac{25}{49}(3, 4)$

1 point

Let  $x = (1, 2)$ , find the projection of  $x$  in a direction perpendicular to  $(3, 4)$ .

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$\frac{25}{49}(-4, 3)$

☐

$\frac{11}{25}(4, -3)$

☐

$(-\frac{26}{49}, -\frac{2}{49})$

☐

$(\frac{26}{49}, -\frac{2}{49})$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(-\frac{26}{49}, -\frac{2}{49})$

1 point

Consider an orthogonal basis  $\{(1, 2, 1), (-2, 0, 2)\}$  of a subspace  $W$ , of the inner product space  $R^3$  with respect to the dot product. If  $y = (1, 2, 3) \in R^3$ , then find  $Proj_W(y)$ .

☐

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$

☐

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$

☐

$18 - 8$

☐

$32, 8$

☐

$36, 6$

☐

$18 + 8$

☐

$32)$

☐

$(\frac{1}{3}, -\frac{8}{3}, \frac{7}{3})$

☐

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, -\frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$

☐

$\sqrt{\phantom{x}}$

$$18 - 8$$

$$\sqrt{\quad}$$

$$32, 8$$

$$\sqrt{\quad}$$

$$36, - 6$$

$$\sqrt{\quad}$$

$$18 + 8$$

$$\sqrt{\quad}$$

$$32)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\left(\frac{1}{3}, \frac{8}{3}, \frac{7}{3}\right)(31, 38, 37)$$

[Check Answers](#)

Your score is 0/6